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A comparative study on the lubricant properties of chemically modified Calophyllum inophyllum oils for bio-lubricant applications

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ABSTRACT

The trend of using non-edible vegetable oils as a potential base stock for industrial lubricants is increasing due to environmental issues and fossil fuel depletion. *Calophyllum inophyllum* oil (CIO) is a non-edible tree seed oil available in various parts of the world. The major disadvantage of CIO is the high acid and peroxide value, which leads to poor oxidation stability. This study focused on the chemical modification of CIO by epoxidation and transesterification processes to improve its oxidation stability. The fatty acid profile of CIO used in the study was obtained using Gas chromatography - Mass spectroscopy technique and the functional groups present in CIO and chemical modified CIO were identified using Fourier transform infrared spectroscopy. The lubricant properties such as tribological, rheological, physicochemical, thermal properties, and oxidative stability of pure CIO, epoxidized CIO (ECIO), and transesterified CIO (TCIO) have been evaluated and compared in this study. The experimental results indicated that there is a significant improvement in the chemical properties and oxidative stability of CIO after the transesterification and epoxidation process. The viscosity and wear scar diameter of ECIO was observed to be better than that of CIO by 42.81% and 7.97%, respectively. TCIO has shown the least viscosity increment ie., 25.45% after 120 h of heating which shows better oxidative stability. The modified CIO can be considered as a potential base-stock for bio-lubricant formulation.

1. Introduction

The use of lubricants will reduce friction between surfaces and are widely used to facilitate easiness in motion. Due to the rising demands of industrial lubricants, vegetable oils are a well-thought-out possible alternative to commercial lubricants. They are biodegradable, innocuous, and are renewable (Boyde, 2002). They also have lower evaporative losses, a broad viscosity range, excellent lubricity, and high flash point. Even if the vegetable oil possesses many advantages, oxidative stability, low-temperature properties, and chemical properties of these vegetable oils are not likely favorable to use them directly as lubricants. These negative aspects of vegetable oil can be addressed by chemical modification.

The cost associated with oil is another problem concerning the industrial sectors. Therefore, the research should have to be initiated to obtain a reasonable profit or margin from these replacements (Peralta Álvarez et al., 2016). So the best sought out alternative is the use of non-edible oils. Since edible oils are not used up for daily activities, the rising demand and prices can be so reduced (Pranav et al., 2021). Non-edible oils can also be cultivated in wastelands to increase production. India has between 40 and 64 million hectares of barren land that can be used in part or entirely for the production and cultivation of non-edible oils (Francis et al., 2005). In this research work, CIO is selected as the base oil for bio-lubricant formulation.

CIO is a better alternative due to its abundance worldwide and is non-edible. *Calophyllum inophyllum* trees are large evergreen trees native to Asia. They are widely seen across many tropical countries (Atabani et al., 2013; Venkanna and Reddy, 2009). It usually reaches an average of around 12 m in height. The fruits of the tree are round, green, and drupe. The ripe fruits show a brownish and wrinkled shape. The nut yields about 70%–75% of dark-greenish oil when cold-pressed. The poultice from leaves, flowers, and oil are widely used in traditional medicines (Bhat et al., 1954; Dweck and Meadows, 2002). It is also used for the cases like rheumatism. The oil is also widely been used in the

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preparation of varnish and was used in lighting lamps (Dwivedi et al., 2011). CIO is very rich in antioxidants and contains UV-absorption characters. The oil is commonly found to be used in the dermatology field to treat wounds and scars. They also possess cytotoxic and antibacterial properties(Léguillier et al., 2015).

Many research works to formulate better lubricants from vegetable oils have been investigated (Syahir et al., 2017; Zainal et al., 2018). Non-edible oils like Neem and Mahua when used as cutting fluids impart better surface roughness and reduced tool wear, respectively, compared with commercial cutting fluid (Puttaswamy and Ramachandra, 2018). Compared to SAE 20W40, Hydnocarpus wightianus oil has a higher flash point and fire point. With 1.5 percent ZDDP, it also improved the friction and wear qualities (Salaji and Jayadas, 2021). Rather than using vegetable oils directly, the addition of additives and modifications improved the lubricant behavior. Moreover, it is required to advance the restricted range of viscosities to make a vegetable oil-based lubricant workable (Bokade and Yadav, 2007). Xu et al. reported that the eco-friendly Pentaerythritol rosin ester synthesized from abietic acid as additives are added to rapeseed and soybean oil to improve oxidative stability and viscosity (Xu et al., 2019). Many other works are also researched in the area of preparing a potential lubricant from vegetable oil (Pranav et al., 2021; E Sneha et al., 2020; Edla Sneha et al., 2020). Transesterification, hydrolysis, modification with fatty amines, fatty amides, fatty alcohols, epoxidation of fatty acids/esters, ring-opening reactions, and polymerization processes generate modified vegetable oils with specific usages in lubrication (Karmakar et al., 2017).

The oil selected for the experimentation, CIO, is having a high acid value which directly causes to reduce oxidative stability of the oil. It was reported that various chemical modifications like transesterification and epoxidation had shown improvement in the oxidative stability of vegetable oil along with other lubricant properties. The use of palm oil methyl ester (POME) as an addition in the application of sliding wear of cast iron on mild steel improved the lubricating performance significantly. POME served as an antiwear additive at a concentration of 5%, lowering the rate of wear to a bare minimum (Masjuki and Maleque, 1996). It was reported that the elimination of unsaturated bonds from the structure of the triglycerides in vegetable oil improved the oxidative stability of epoxidized oils considerably (Wu et al., 2000). Transesterification of canola oil with various alcohol combinations using a batch type of reactor provided the oil with better properties as per ASTM standards, of which Ethyl ester showed maximum lubricity (Kulkarni et al., 2007). Transesterification of jatropha oil using ethanol and sodium hydroxide with trimethylolpropane (TMP) has shown improvement in lubricant properties (Resul et al., 2012). Hexagonal boron nitride as additives on modified jatropha oil by a two-step acid-base catalyzed transesterification process using methanol and sulphuric acid in the first stage and methanol and sodium hydroxide in the second stage followed by the addition of TMP in methyl esters provides better tribological properties, improved tool life, and decreased cutting forces (Talib and Rahim, 2018). Candida antartica lipase was used as a catalyst in a solvent-free system to transesterify the reaction between palm stearin and methanol. The reaction settings were modified and the lubricant has shown better properties to be utilized as a base stock because of its better biodegradability and tribological characteristics (Afifah et al., 2019). Chemical modification of potassium hydroxide-based base-transesterification using methanol on castor, palm kernel, and coconut oils impart better quality to being a motor engine-based bio-lubricant (Tulashie and Kotoka, 2020).

Epoxidized soybean oil using alkylated phenolic compound outperformed pure soybean oil in terms of thermal and wear characteristics. The cross-linking of epoxide rings on the surface of metals, which decreases friction between moving components produced a reduction in friction and wear (Adhvaryu and Erhan, 2002). For the production of peracetic acid, acetic acid was combined in equal parts with hydrogen peroxide and sulphuric acid was used as a catalyst. Later this peracetic acid was added to the respective oils for the sal oil and caper spurge oil epoxidation processes. These respective oils were found to have higher viscosities (Chaurasia et al., 2020). When compared to pure rice bran oil, epoxidized rice bran oil using hydrogen peroxide and glacial acetic acid showed a considerable improvement in tribological and lubricating performance. This is because, the oxirane rings on the metal surfaces were cross-linked and hence the tribological properties of epoxidized RBO were observed to be better (Thampi et al., 2020a). Formulation of Linoleic acid-based hydroxytetrahydrofuran ester bio-lubricant using selective chemo-enzymatic epoxidation has shown excellent properties comparing conventional lubricants. The viscosity, viscosity index, pour point, thermal stability, and physicochemical properties revealed that the properties are finest with better performance (Xu et al., 2021). The synthetic bio-lubricant base stocks appear to be suitable for a hydraulic fluid application after a series of saponification and hydrolysis, urea crystallization, epoxidation, epoxide ring-opening, and esterification step chemical reactions and modifications to convert triglyceride of jatropha and soybean oils to bio-lubricant (Pindit et al., 2021). Epoxidation was carried out in caper spurge oil using acetic acid and hydrogen peroxide and ZnO nanoparticles were added to improve the frictional behavior. The coefficient of friction (COF) of epoxidized oil has considerably reduced when compared to pure oil (Singh et al., 2021).

The methyl ester from CIO seed meets the standards set by the United States (ASTM D 6751–06) and European Union (EN 14214) (Azam et al., 2005). The biodiesel production involves removing the glycerol molecules as a by-product, and the purification of esters formed (Fernando et al., 2007). Biodiesel prepared from this oil is found to be in a close range to diesel engine requirements (Sahoo et al., 2009). Two-stage transesterification is usually performed for CIO biodiesel preparation (Habibullah et al., 2015b). The reaction of TMP with CIO methyl ester in a batch stirred reactor yielded CIO-based TMP ester. In comparison to paraffin mineral oil and commercial lubricant, CIO-TMP ester had the lowest coefficient of friction at all test temperatures. This is due to the fatty acid's ability to form a boundary film, which increases lubricity (Habibullah et al., 2015a). Compared to other biodiesel fuels, biodiesel prepared from CIO has shown more significant potential. Results are comparable, and the blends with diesel reduced NOx emissions significantly (Fadhlullah et al., 2015; Vigneshwar et al., 2019). There was a considerable reduction of COF and wear scar diameter when 5% and 10% CIO biodiesel was blended with commercial oil (Zaid et al., 2020). These previous studies have indicated that CIO has shown better properties as a bio-fuel, but very few research studies have been carried out on CIO as a bio-lubricant. This research paper focuses on the lubricant property comparison of bio-lubricant formulated from CIO using epoxidation and transesterification processes which are not yet revealed and reported. Thus the scope is a very novel approach to consider. These types of studies are well advantageous to reduce the usage and concerns regarding uncontrolled conventional lubricants, thereby introducing sustainability in production.

In the present study, chemical modifications such as transesterification and epoxidation have been carried out to improve the chemical and oxidative stability of the CIO. The fatty acid composition of the pure base oil has been evaluated initially. The chemical modification process was confirmed by performing an iodine value test and Fourier transform infrared (FTIR) spectroscopy. The major lubricant properties such as tribological, rheological, physicochemical, thermal, and oxidative stabilities of CIO, ECIO, and TCIO have been evaluated and compared in this paper. The chemically modified CIO has shown the ability to be used as a base-stock for bio-lubricants.

2. Materials and methods

2.1. Materials

Calophyllum Inophyllum oil, locally called Tamanu/Punnai/Polanga oil is extracted from the nuts of Calophyllum Inophyllum trees. The nuts collected were cold-pressed to extract the oil. The base oil used in the

study was supplied from Krishna Ayurvedics, Thiruvananthapuram, Kerala, India. The chemicals such as acetic acid, 30% hydrogen peroxide, sulphuric acid, and potassium hydroxide (KOH) were purchased from Nice chemicals, Kochi, Kerala, India. Methanol was purchased from Molychem, Mumbai, India. Trimethylolpropane and sodium methoxide were purchased from Otto chemicals, Mumbai, India.

2.2. Fatty acid profile

The main content of vegetable oil is the fatty acids present in it. These fatty acids are hydrocarbon compounds that will influence the lubricant properties of base oil based on the carbon chain length and the presence of double bonds. The fatty acid profile of the same oil may differ due to topographical changes across areas. So, it is important to evaluate the fatty acid content of the base oil before formulating a bio-lubricant. The fatty acid profile of crude CIO used in this study was evaluated using the Gas chromatography and Mass spectroscopy (GC-MS) technique - Agilent Technology (USA) model 7890A-5975C.

2.3. Chemical modification- epoxidation and transesterification

The epoxidation of the CIO was performed as mentioned below and the chemical reaction is shown in Fig. 1. Initially, 5 ml acetic acid and 3–4 drops of sulphuric acid were added to 100 ml of CIO. Then the mixture was swirled manually and 20 ml of hydrogen peroxide taken in a burette was added dropwise to the mixture. The mixture was then constantly magnetic stirred with heating in a range of 65°C–75 °C. After 4.5 h, the mixture was poured into a separating funnel and was kept undisturbed. After the separation, remove the clear liquid from the bottom. The oil obtained was then water-washed with hot deionized water. The temperature of deionized water was kept at around 70 °C. The final mixture obtained was then heated using an electric heater with a vacuum pump to remove the excess water in the oil sample.

Transesterification of CIO is a two-stage process due to the higher content of free fatty acids present. The transesterification of the CIO was performed as mentioned below and the chemical reactions are shown in Fig. 2 and Fig. 3. Initially, 2 ml of sulphuric acid and 64 ml of methanol were poured into a 200g CIO sample. The mixture was then constantly stirred in a water bath at 75 °C. After 2 h, the mixture was poured into a separating funnel and was kept undisturbed. After 6 h, the unreacted sulphuric acid layer formed on top of the separating funnel was removed. The remaining oil was then added with a mixture of 2g of KOH and 55 ml of methanol. The mixture was again constantly stirred in a water bath at 65 °C. After 1 h, the mixture was poured into a separating funnel and kept undisturbed. After 6 h, the glycerol formed at the bottom of the separating funnel was removed, and the mixture was water-washed with hot deionized water. The temperature of deionized water was kept at around 70 °C. Then the mixture was heated using an electrical heater with a vacuum pump to remove the excess water in the oil. The two-stage transesterified oil was then modified by adding Trimethylolpropane in a 1:4 ratio with the modified oil. 1% sodium methoxide was added as a catalyst. The mixture was constantly stirred at a temperature of 80 °C for 2 h, and the final product was obtained after filtration. The experimental setups for chemical modification are shown in Fig. 4.

2.4. Physico-chemical properties

The physical properties like the color and density of the oil samples were noted. The density of the oil samples was evaluated using a specific gravity bottle. The chemical properties like acid value, peroxide value, and iodine value of all the samples have been evaluated using the titration method as per IS: 548 (Part 1) – 1964. The acid value is the amount of base required to neutralize lubricant. That is, it gives the free fatty acid content present in the oil. The peroxide value is a parameter specifying the content of peroxide, especially hydroperoxides, in the oil. It is a primary source of oxidation. The critical sites such as β-carbon, unsaturated bonds, present in the oils are the sites of oxidation, which can worsen the oil by making it rancid. Both the acid and peroxide values will give an indirect measure of oxidation. The iodine value gives the measure of unsaturation present in the samples and is expressed as the amount of iodine absorbed per gram of oil.

2.5. Fourier transform infrared spectroscopy

The FTIR spectroscopy was used to evaluate the functional groups of base oil and chemically modified oil samples using Thermoscientific Nicolet iS50 with a data recording software. The spectrum for all the oil samples was obtained in the range of 400–4000 cm^{-1}.

2.6. Rheological properties

The kinematic viscosities of CIO, ECIO, and TCIO were evaluated in this study using Cannon-Fenske Opaque Viscometer as per the ASTM D446 standard. For various industrial lubricants, the viscosity should be optimal for its application.

2.7. Thermal properties

The Low-temperature properties like pour and cloud points of the samples have been evaluated as per ASTM D-97, and ASTM D-2500, respectively. These values are more critical during the winter season and while using lubricants in cold counties. If the cloud and pour points of the lubricants are not good, the lubricity will be more affected. Upon long-term exposure to cold temperature, vegetable oils tend to form macro-crystalline structures due to the bend happening in the triglyceride structure. These structures even self-stacks and lose their kinetic energy (Erhan et al., 2006). So, a good lubricant should have sufficiently lower cloud and pour points.

The Thermogravimetric analysis (TGA) determines the sample's weight change as a function of a temperature profile. In the present work, this approach is used to determine the decomposition of the oil samples over a temperature range of room temperature to 800 °C. The test was carried out in an oxygen atmosphere using a simultaneous thermal analyzer- STA 8000. The temperature at which mass degradation started and complete degradation happened was also observed. So the oil sample having lower thermal stability will decompose first and result in sudden mass degradation.

2.8. Tribological properties

The tribological properties such as COF and wear scar diameter

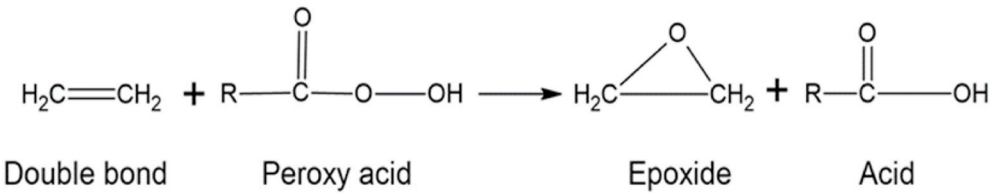


Fig. 1. Epoxidation reaction.

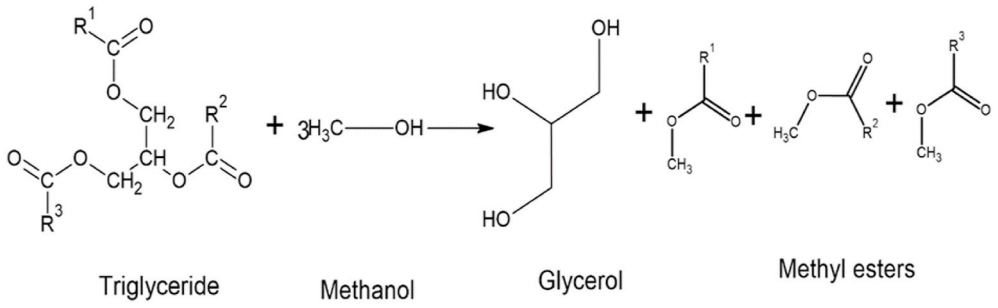


Fig. 2. Formation of methyl esters (Trasnesterification- First stage).

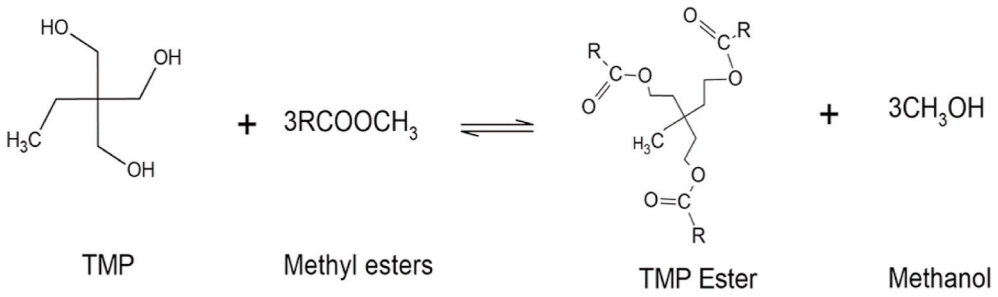


Fig. 3. Formation of TMP esters (Transesterification- Second stage).

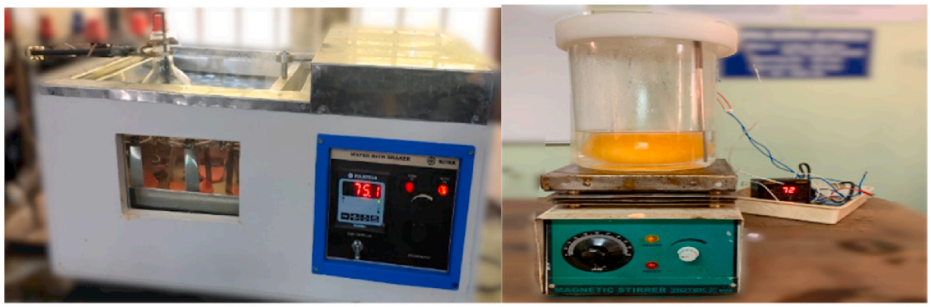


Fig. 4. Chemical modification setups for (a)transesterification and (b)epoxidation.

(WSD) are evaluated as per the ASTM D-4172 standards. These properties are very important to elucidate the performance of a good lubricant. The tribological properties for all the samples are performed on the four-ball tribo-tester. Chrome alloy steel balls with 12.7 mm diameter and 61 HRC are used in testing. The tribometer is run at 75 °C, 1200 rpm, 392 N load is applied for 3600 s. In the test apparatus, the load is applied using a pneumatic system, and the load applied is measured using a load cell connected to the machine. During spindle rotation, a frictional force is created at the point of contact of balls, and this force is measured using another load cell and displayed as frictional torque after multiplying with the torque distance. The COF is obtained from the frictional torque and load applied using the equation:

$$Frictional\ torque\ (Nm) = \mu \times N \times r \quad \text{eq. (1)}$$

where, μ = COF, N = resultant force = load applied/cos (35.2644), and r = distance on contact of ball from center = 0.00366m.

The WSD of the test balls has been observed using an optical microscope attached along with the machine. The cleaned bottom chrome alloy balls kept stationary in the ball pot will be taken for evaluation of the wear scar. The balls will be placed beneath the microscope and the average wear scar value of three balls is recorded and reported. The worn surface images caused by the test samples are obtained with the help of a Leica optical microscope.

2.9. Flash temperature parameter

This Flash temperature parameter (FTP) is a value that determines at what temperature the lubricant film breaks down. This temperature will predict the safe operating condition of the particular lubricant. If the FTP is low, that lubricant is having poor characteristics at higher temperatures (Rahman et al., 2017; Singh, 2015). The FTP was calculated using the eq. (2) where d = average wear scar diameter in mm and W = applied load in kg.

$$FTP = \frac{W}{d^{1.4}} \quad \text{eq. (2)}$$

2.10. Oxidative stability

The vegetable oils have low oxidative stability because of polyunsaturated acids like linoleic and linolenic acids (Wu et al., 2000). Therefore, the hot oil oxidation test (HOOT) as per the American Oil Chemists Society (AOCS Cd-12-57) was performed in this study to evaluate the oxidative stability of the oil samples. The experiment was conducted in a hot air oven at 100 °C for 120 h to accelerate oxidation.

Due to the formation of oxidative bi-products, the viscosity of the oil samples increases; hence the viscosities of all the oil samples were examined every 24 h at 40 °C. After 120 h, the sample which has the lowest percentage of viscosity increment will have better oxidative stability.

3. Results and discussions

The fatty acid composition of CIO obtained from the chromatogram shown in Fig. 5 using the GC-MS technique is listed in Table 1. The area under the peak shown in the chromatogram is proportional to the percentage amount of the compound. A total of six peaks have been observed in Fig. 5 which constitutes six different compounds. In order to identify the various compounds, the retention time of each compound is compared with the retention time of standard compounds (Nair et al., 2017).

From Table 1, it was noted that the major compounds present in CIO are stearic acid (43.33%) and linoleic acid (31.18%). The total saturated and unsaturated acids are 58.77% and 41.24% respectively. Saturated acids induce better oxidation stability, and unsaturated acids impart better low-temperature properties. The monounsaturated fatty acids are less vulnerable to oxidation than polyunsaturated fatty acids. Unsaturation within the chemical structure might be a favorable place for oxidation to occur (Nogales-Delgado et al., 2021). Since CIO contain both saturated and unsaturated fatty acids in almost equal proportion, an optimal level of oxidation stability and low-temperature properties can be achieved. Even though CIO can't be directly used as a bio-lubricant, chemical modifications/suitable additives are necessary for attaining favorable lubricant properties (Fadhlullah et al., 2015; Ranjithkumar et al., 2021; Zainal et al., 2018).

The Physico-chemical properties like density, color, acid, peroxide, and iodine values were evaluated and tabulated in Table 2. Chemical modifications like transesterification and epoxidation have improved the chemical properties of the oil. The acid value which is the measure of free fatty acids was 36.5 mg KOH/g for the pure CIO. After the chemical treatments, it was observable that the acid values of TCIO and ECIO were reduced by 95.56% and 22.08%, respectively to CIO. The reduction in acid value indicates a better shelf-life of modified oils. Peroxide values of the TCIO and ECIO samples have also been reduced by 95.87% and 92.51%, respectively compared to the pure oil. The considerable decrement of peroxide value is very much favorable to use it as a lubricant since the primary oxidation was inhibited by eliminating the critical oxidation sites, thereby increasing the oxidative stability. The iodine number determined by titration is an easy way to validate the presence of remaining double bonds in fatty acid chains (de Haro et al., 2016). It was observable that the iodine values of TCIO and ECIO have been reduced by 12.66%, and 23.85%, respectively, with respect to the pure CIO. The epoxidation reaction removed the unsaturated bonds which resulted in the significant reduction of iodine value in ECIO. Generally, these reductions in chemical properties of the modified oils indicate better oxidative stability and shelf-life.

Table 1
Fatty acid composition of CIO.

Name of acid	Structure	Nature	Content (%)
Palmitic Acid	C16:0	Saturated	14.69
Stearic Acid	C18:0	Saturated	43.33
Eicosanoic Acid	C20:0	Saturated	0.75
Oleic Acid	C18:1	Mono unsaturated	7.9
Linoleic Acid	C18:2	Poly unsaturated	31.18
Linolenic Acid	C18:3	Poly unsaturated	2.15
Total Saturated acid			58.77
Total Unsaturated acid			41.23
Total Polyunsaturated acid			33.33

Table 2
Physico-chemical properties of CIO and its derivatives.

Oil sample	Chemical properties			Physical properties	
	Acid value (mg KOH/g)	Peroxide value (Meq/kg)	Iodine value (gI_2/100g)	Density (g/cm^3)	Color
CIO	36.50	5.57	93.92	0.876	Dark greenish
TCIO	1.62	0.23	82.03	0.65	Reddish orange
ECIO	28.44	0.42	71.52	0.902	Greenish-yellow

Comparing the density of the oils, ECIO has shown an increased density, on the other hand, for TCIO, density was reduced. Since density is dependent on the molecular weight of the samples (Rodenbush et al., 1999), ECIO has higher molecular weight compounds (epoxides) formed when compared with TCIO which possesses lower molecular compounds like TMP-esters. After transesterification, the density of the oil is reduced due to the replacement of glycerol with TMP which is having a lower density.

The CIO, ECIO, and TCIO samples are shown in Fig. 6.

The FTIR spectroscopy of all the oil samples has been recorded and the spectrums of pure and modified oils are shown in Figs. 7 and 8. Fig. 7 shows the FTIR spectrum of the CIO and ECIO. The major spectrums are identified at 1237 cm^{-1} and 818 cm^{-1} peaks due to the formation of epoxy and the epoxide group after the epoxidation reaction. Other important peaks are observed at 2921 cm^{-1}, 1740 cm^{-1}, and 1455 cm^{-1},

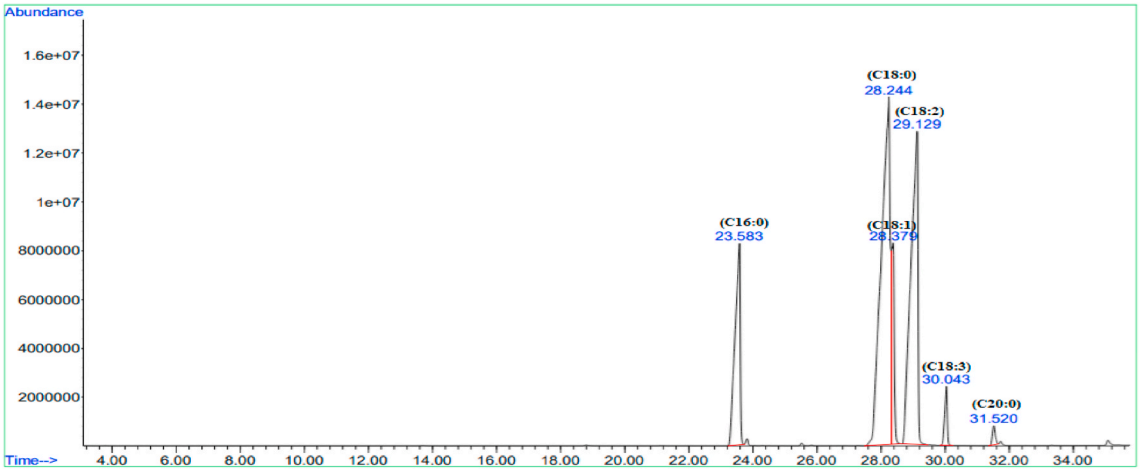


Fig. 5. GC-MS chromatogram of CIO.

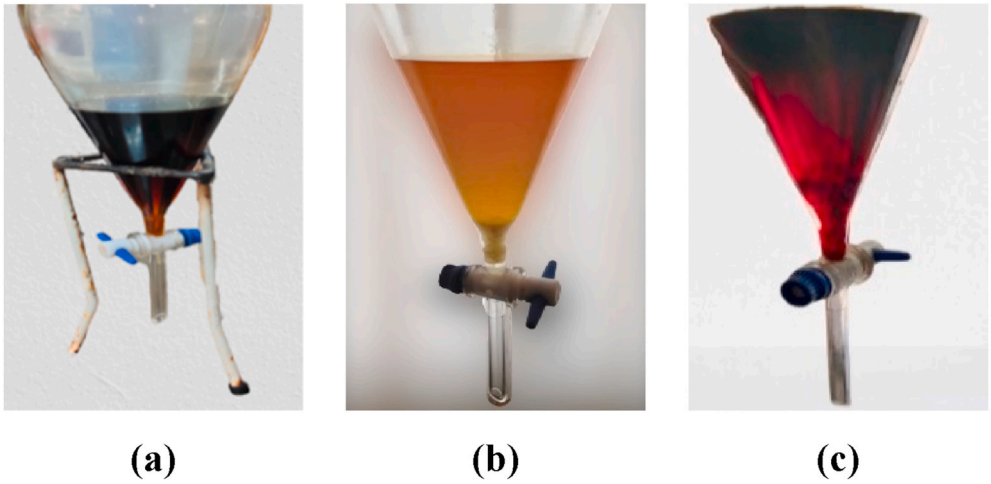


Fig. 6. (a) CIO (b) ECIO (c) TCIO

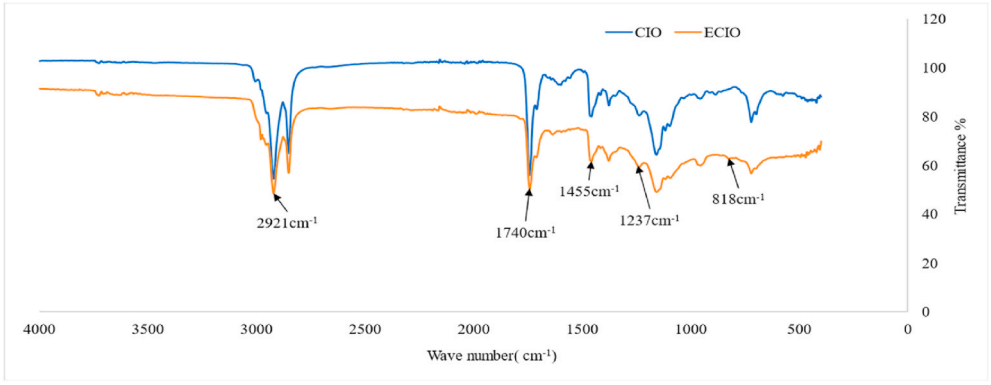


Fig. 7. FTIR results of CIO and ECIO.

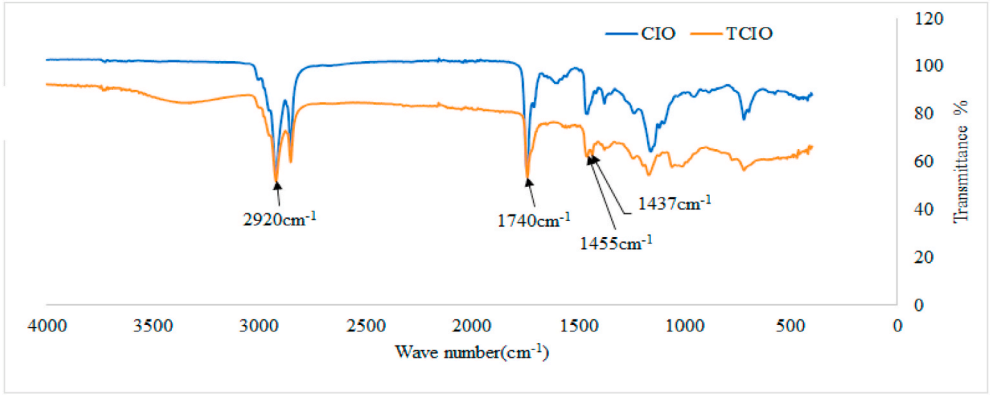


Fig. 8. FTIR results of CIO and TCIO.

corresponding to the alkene, ester, and alkane, respectively.

Fig. 8 shows the FTIR spectrum of the CIO and TCIO. The major spectrums are identified at the 1437 cm^{-1} peak, which was due to the formation of the CH_3–O group during the first stage of the reaction. Other significant peaks are observed at 2920 cm^{-1}, 1740 cm^{-1}, and 1455 cm^{-1}, corresponding to the alkene, ester, and alkane, respectively.

The kinematic viscosities of CIO, TCIO, and ECIO are shown in Fig. 9. When it comes to viscosity, it is one of the most essential features of lubricating oil that defines its application and performance under operating circumstances (Kania et al., 2015). Generally, the metal cutting operation utilizes lubricants with lower viscosities. Oils with higher viscosity are better to be used as engine oils and in high-temperature applications (Ting and Chen, 2011). The reduction in unsaturation generally improves the viscosity (Rodrigues et al., 2006). The presence of a longer alkyl chain and the branching of compounds will also improve the viscosity of the products (Cermak and Isbell, 2009). In epoxidation a similar trend was observed that the carbon-carbon double bonds are converted to an epoxide ring-like structure, which significantly increased the saturation content in terms of iodine value, thereby increasing the viscosity. But the formed epoxide

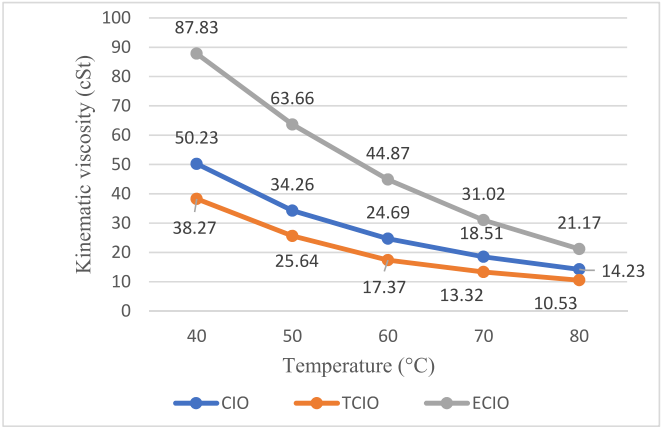


Fig. 9. Variation of kinematic viscosity of oil samples with respect to temperature.

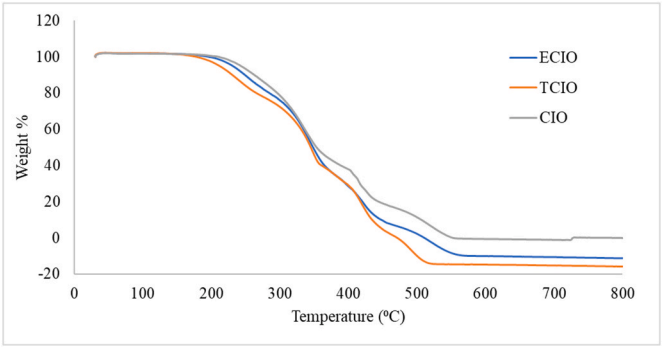


Fig. 10. TGA curve of CIO and its derivatives.

rings are unstable at a higher temperature reducing the viscosity drastically with an increase in temperature. In transesterification reaction, the high viscous glycerol part was separated from the structure of the triglycerides, hence a reduction in viscosity was observed in TCIO even after a slight reduction in unsaturation.

Low-temperature properties like cloud and pour point have been evaluated and are tabulated in Table 3. The saturated acid tends to close pack at lower temperatures making the oil to get solidify easily ie., the removal of unsaturated bonds causes a negative impact on low-temperature properties (Jayadas and Nair, 2006). However, introducing a branched-chain in fatty acids will improve the low-temperature properties by preventing molecular packing under lower temperatures (Erhan et al., 2006; Jayadas and Nair, 2006). In chemically modified CIO, the iodine value was noted to be lower which indicates the presence of less amount of unsaturation. Whereas the FTIR results indicated the presence of epoxy and epoxide groups in ECIO, and the CH_3–O group in TCIO maintained the low-temperature properties even after chemical modification.

The TGA of all the oil samples is shown in Fig. 10. TGA is widely used to evaluate the thermal stability of synthesized samples and the sudden mass degradation of the sample indicates the thermal instability (Duangkaewmanee and Petsom, 2011; Fox and Stachowiak, 2007). The thermal degradation of the samples is shown as a percentage change of weight with respect to the temperature of the oil. From Fig. 10, it was observed that all the oil samples show comparable thermal stability before and after chemical modifications, even though a slightly better property was seen in CIO. It was noted that the mass degradation initially started at 196.43 °C, 178.55 °C, 210.86 °C for ECIO, TCIO, and CIO respectively. Whereas the complete degradation of CIO was observed at 552.77 °C, ECIO at 510.83 °C, and TCIO degraded at 472.22 °C.

The Tribological properties such as average wear scar diameter and coefficient of friction obtained for CIO, TCIO, and ECIO are tabulated in Table 4. The glycerol part present in the vegetable oil tends to adhere to the metallic surface. This is due to the better metal affinity and stability of lubricant films of pure vegetable oils (Rudnick, 2020). Since these glycerols are removed after transesterification, COF and wear scar diameter are seen to increase (Zainal et al., 2018). The microscopic images of the worn surfaces are shown in Fig. 11. It was noted that the

Table 3
Cloud and Pour point of CIO and its derivatives.

Oil Sample	Cloud point (°C)	Pour point (°C)
CIO	10	9
TCIO	10	8
ECIO	10	8

Table 4
Tribological properties of CIO and its derivatives.

Oil Sample	COF	WSD (μm)
CIO	0.0321	577
TCIO	0.0383	664
ECIO	0.0381	531

wear scar diameter was reduced after epoxidation of CIO due to the formation of higher molecular compounds such as epoxides along with the presence of glycerol part. These oxirane rings formed might have adhered to the metal surfaces which resulted in reducing the WSD (Thampi et al., 2020a). This property of the bio-lubricant to reduce friction and wear with minimal surface undulations is highly significant. The WSD image of CIO is shown in Fig. 11(a). It was observed that the worn surface caused by CIO contains scratches and welded material removal sites that occur because of the abrasive and adhesive wear respectively. The wear surface of CIO is more characterized to have deep grooves and undulations comparing TCIO and ECIO. Free fatty acids present in CIO produce metallic soap coatings caused by chemical reactions, which are continually eliminated by sliding motion. As a result, deep grooves are produced. The same phenomenon is observed in other studies also (Gul et al., 2020; Jayadas et al., 2007; Mannekote and Kailas, 2012). Also, it was observed that the worn surfaces of ECIO and TCIO shown in Fig. 11(b) and (c) exhibited only abrasive wear. The wear grooves formed by ECIO was observed to be smooth with less undulations.

The FTP of all the selected oils has been calculated using the eq. (1) and are tabulated in Table 5. The load applied is taken as 40 kg. When the tribofilm formed between the surfaces is better, the wear scar diameter will be lower and if a poor tribofilm is formed, it will create higher wear scars (Thampi et al., 2020b). In these modified oils, ECIO is having the least wear scar diameter comparing other oils, hence is having a better FTP.

The oxidative stabilities of all the modified oil samples have been evaluated and compared with pure CIO, and the results are tabulated in Table 6. The oxidative stabilities of the oils were measured with respect to the change in kinematic viscosities since the rate of oxidation increases the viscosity of oils. The kinematics viscosities of pure and modified CIOs have been evaluated after HOOT.

The higher percentage increment of viscosity in pure CIO was observed because of the presence of high free fatty acid content and highly reactive oxidation sites. These sites will promote oxidation, and it will increase the viscosity by the formation of higher molecular compounds like hydroperoxides, peroxides, aldehydes, ketones, etc. which eventually results in agglomeration, sedimentation, and sludge formation. The percentage increment of the viscosities of CIO, TCIO, and ECIO after 120 h of HOOT are observed to be 77.02%, 25.45%, and 35.04%, respectively. The least viscosity increment was observed for TCIO this might be because of the lower acid and peroxide values i.e., 1.64 mg

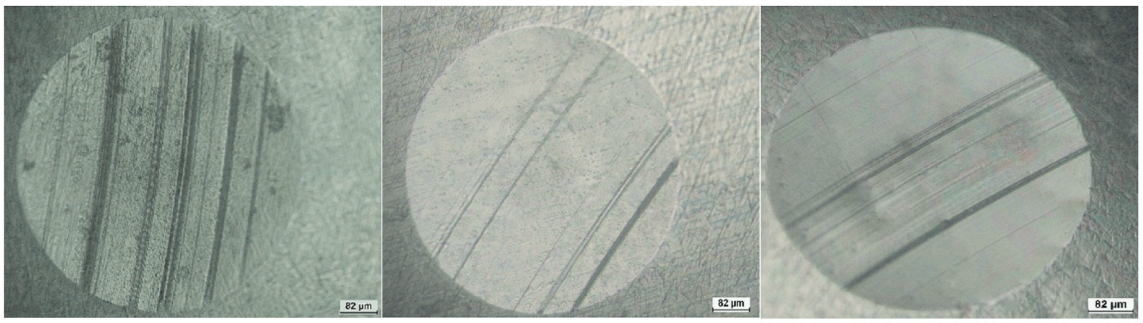


Fig. 11. Worn surface image (10 ×) from microscope (a)CIO (b)ECIO (c)TCIO.

Table 5
FTP value of CIO and its derivatives.

Oil Sample	FTP
CIO	86.38
TCIO	70.96
ECIO	97.03

KOH/g and 0.23 Meq/kg respectively. These values are very low in comparison with the acid and peroxide values of pure CIO and ECIO, hence it has shown better oxidative stability. The hydrogen molecules on the beta carbon position in the glycerol part were removed during the transesterification process and replaced with TMP, which largely improved the oxidative stability of the TMP ester (Yunus et al., 2004). Comparing pure CIO and ECIO, ECIO has shown better oxidative stability due to the lower acid, peroxide, and iodine values compared to CIO. From Table 6, it is clear that TCIO and ECIO have shown better oxidative stability compared to CIO in terms of viscosity increment.

4. Conclusion

Vegetable oils show better potential to perform as an alternative to commercial lubricants. The oil selected for the study CIO is a non-edible oil. Since this oil is not used in daily activities, the competition with respect to edible oils can be eliminated. Based on the experimental investigations, the following conclusions have been observed-

- The fatty acid profile of the CIO has shown the presence of a higher amount of stearic acid (43.33%). Saturated acids constitute 58.77% and unsaturated acids are about 41.23% in the composition.
- One of the major drawbacks of CIO is its lower oxidative stability and high acid value. Chemical modification processes such as transesterification and epoxidation have substantially reduced these drawbacks.
- After chemical modifications, the free fatty acid content in transesterified and epoxidized samples has been reduced by 95.56% and 22.08%, respectively, when compared with pure CIO.
- Other chemical properties like peroxide and iodine values have also been reduced after chemical modification. The peroxide values of ECIO and TCIO have substantially reduced and the iodine value is noted to be lowest for the ECIO sample.
- The viscosity of ECIO got enhanced by 42.81% at 40 °C in comparison to CIO. However, the viscosity of TCIO got reduced by 23.81%.
- TGA curve of oil samples showed that both the pure as well as modified samples are having comparable thermal stability at higher temperatures. Whereas, the cold flow properties need to be improved by using a suitable pour point depressant.
- Promising results are also seen in tribological properties. The WSD of ECIO is noted to be the lowest among other oil samples with less wear found in the worn surface. The wear grooves formed by ECIO are also observed to be smooth. The FTP which depends on WSD is found to be better for ECIO.
- The oxidative stability of the modified oils is found to be better. TCIO has shown the least viscosity increment after HOOT followed by ECIO and CIO.
- Comparing transesterification and epoxidation, transesterification is found to be effective in CIO due to the higher amount of free fatty acid content but lower WSD with a smooth worn surface is noted for the epoxidized sample. A suitable combination of multistage chemical modification might provide better lubricant properties which need to be further evaluated as a future scope.

CRediT authorship contribution statement

Pranav Prasannakumar: Conceptualization, Investigation, Methodology, Writing – original draft. **Sneha Edla:** Conceptualization, Investigation, Methodology. **Ananthan D. Thampi:** Investigation, Methodology, Writing – original draft. **Muhammed Arif:** Supervision, Writing – review & editing. **Rani Santhakumari:** Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

Table 6
Oxidative stability of CIO and its derivatives.

Time (h)	CIO		TCIO		ECIO	
	Viscosity (cSt)	% increment	Viscosity (cSt)	% increment	Viscosity (cSt)	% increment
0	50.23	–	38.27	–	87.83	–
24	54.06	7.63	40.43	5.64	95.92	9.21
48	59.42	18.3	42.22	10.32	102.91	17.17
72	68.57	36.61	43.89	14.68	108.15	23.14
96	78.75	56.51	45.08	17.79	113.89	29.67
120	88.92	77.02	48.01	25.45	118.61	35.04

the work reported in this paper.

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